

CHEATSHEET · QUICK REFERENCE

Lab 7 — Partition and Volume Types, File Systems

Compare MBR vs. GPT limits, format ext4/XFS/ZFS, map the five exam filesystems and persist mounts by UUID in /etc/fstab.

Partition tables

```
parted -s /dev/loop10 mklabel msdos
```

Create an MBR table

```
parted -s /dev/loop10 mklabel gpt
```

Create a GPT table

```
parted /dev/loop10 print
```

Show partitions

```
gdisk -l /dev/loop10
```

Show GPT detail / backup header

MBR vs. GPT limits

Attribute	MBR	GPT
Max disk	2 TiB	9.4 ZB
Max partitions	4 primary (or 3+1 ext)	128 default
Redundancy	None	Backup header at end
Boot firmware	Legacy BIOS	UEFI

Format and persist

```
mkfs.ext4 /dev/loop10p1
```

ext4 - Linux default

```
mkfs.xfs /dev/loop10p2
```

XFS - large files

```
zpool create -f tank /dev/loop10p3
```

ZFS pool

```
blkid -s UUID -o value /dev/loop10p1
```

Get the UUID

```
echo "UUID=... /mnt/p1 ext4 defaults,noatime 0 2" >> /etc/fstab
```

Persist by UUID

Filesystem comparison

FS	Native OS	Use case
ext4	Linux	General-purpose servers
XFS	Linux (RHEL)	Large files, online grow only
NTFS	Windows	Windows volumes, NTFS ACLs
ReFS	Windows Server	Storage Spaces, integrity
VMFS	VMware ESXi	Clustered datastore for vmdk
ZFS	Linux/BSD/Solaris	Pooled, checksums, snapshots